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MINISTRY OF EDUCATION, SINGAPORE
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CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Normal (Academic) Level

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MATHEMATICS (SYLLABUS A)

Paper 1

September/October 2025

4045/01

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE ON ANY BARCODES.

Answer all the questions.

The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142.

This document consists of 18 printed pages and 2 blank pages.



Singapore Examinations and Assessment Board



Cambridge Assessment
International Education

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[Turn over



Mathematical Formulae

Compound interest

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi rl$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2}$$





1

Write these numbers in order, starting with the smallest.

- $\frac{5}{4}$
- $\sqrt{3}$
- $\frac{2}{\pi}$
- $\frac{4}{64}$
- 78%

Answer all the questions.

3



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2

Aisha chooses a t-shirt, a pair of shorts and a pair of socks from the following options.

T-shirt
red (R)
green (G)
yellow (Y)

shorts
black (B)
white (W)

socks
striped (S)
plain (P)

Complete the following table to show all her possible choices.
You may not need to use all the rows in the table.

T-shirt	R	R	R											
shorts	B	B	W											
socks	S	P	S											

[2]



Answer

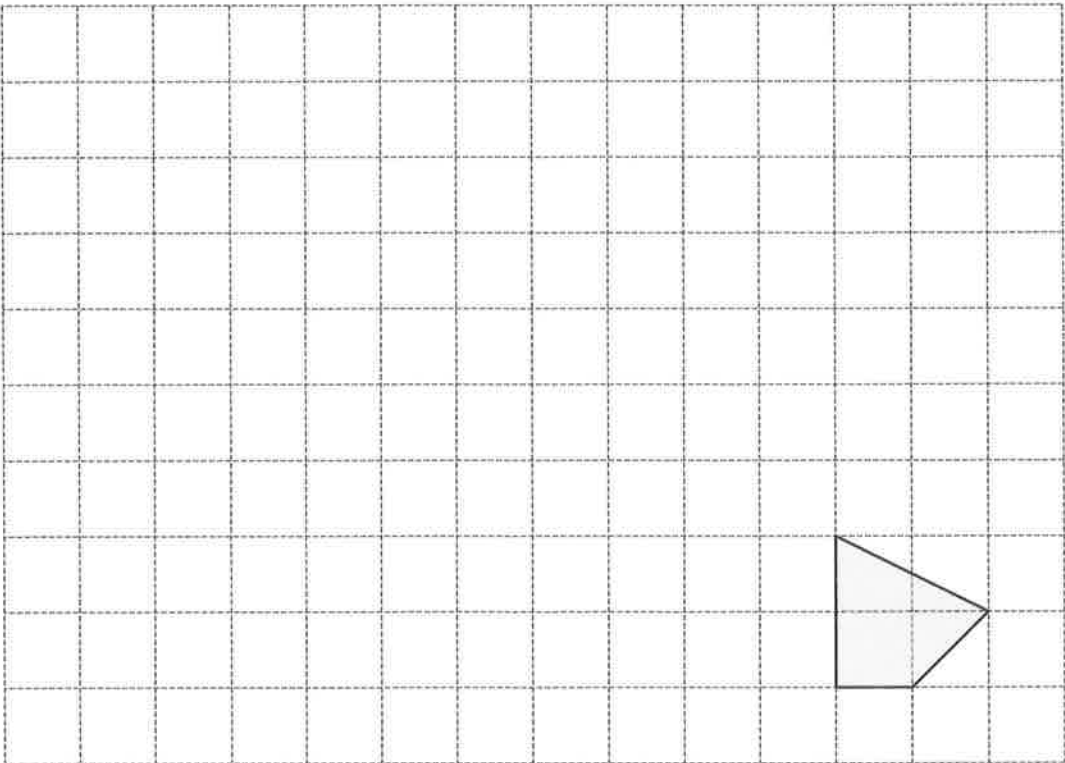
Show that she is **not** correct.

(b) An expression for the n th term of a sequence is $2n^2 - 25$.
Heng says that the value of the third term in this sequence is 11.

Answer,

Write down the next two numbers in this sequence.

4 (a) 16 10 4 -2 ...



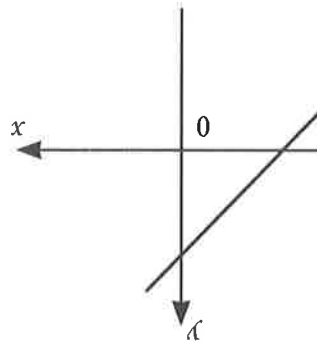
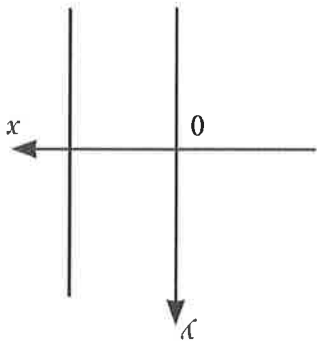
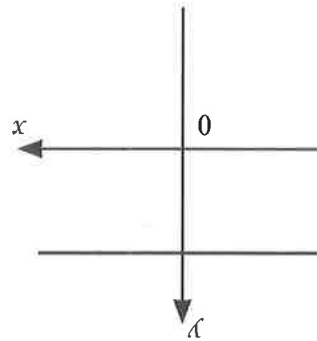
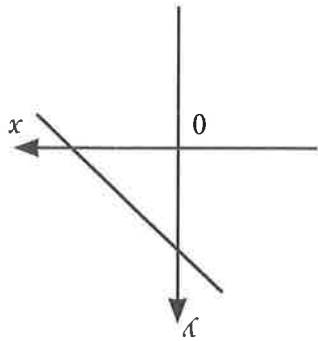
3 Draw an enlargement of this quadrilateral using the scale factor of 3.



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[2]

*Answer*

Write the correct equation below each sketch.

Sketches of the graphs of some of these equations are drawn below.

F $y = -x + 5$

E $y = 5x$

D $y = x + 5$

C $y = 5$

B $y = x - 5$

A $x = 5$





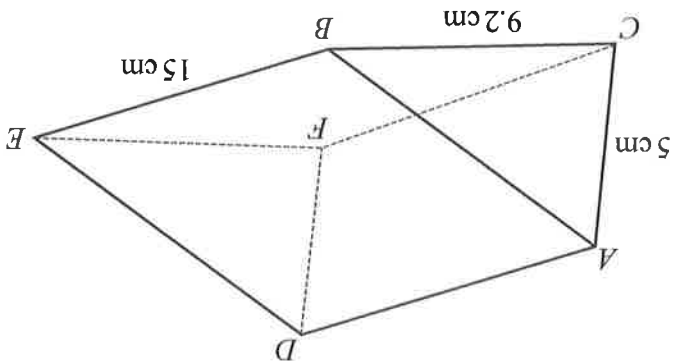
6 Priya travels at a speed of 48 km/h.
Yash travels at a speed of 13 m/s.

Who travels at a faster speed?
Show your working.

Answer

[2]

7



The diagram shows a prism, $ABCDEF$.

Angle ACB is acute.

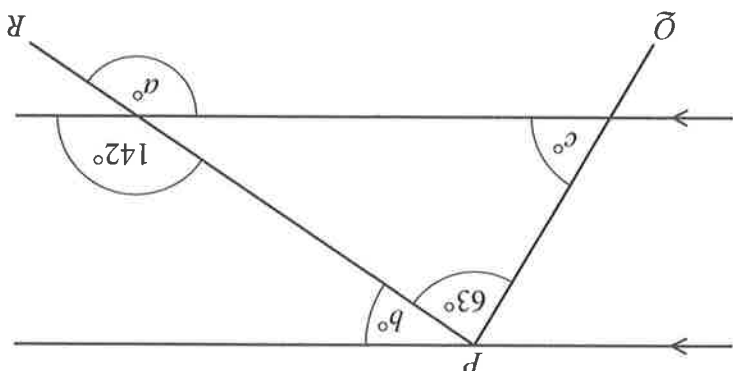
Lim says that the volume of the prism is exactly 345 cm^3 .

Show that he is **not** correct.

Answer

[2]



(c) Find c .Answer $b = \dots\dots\dots$ [1](b) Find b .Answer $a = \dots\dots\dots$ [1](a) Find a . PQ and PR are straight lines that intersect two parallel lines.

9

Answer $\dots\dots\dots$ [2]8 Write $\frac{4x}{3} - \frac{5(x+1)}{4}$ as a single fraction in its simplest form.

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10 (a) Write 45 : 390 as a ratio in its simplest form.

Answer : [1]

(b) The total cost of 6 identical cakes is \$2.52 .

Calculate the total cost of 10 of these cakes.

Answer \$ [2]

11 $720 = 2^4 \times 3^2 \times 5$

(a) Write 720^2 as the product of its prime factors.

Answer [1]

(b) The highest common factor (HCF) of 720 and N is $2^4 \times 3$.
The lowest common multiple (LCM) of 720 and N is $2^7 \times 3^2 \times 5 \times 7^2$.

Write N as the product of its prime factors.

Answer [2]

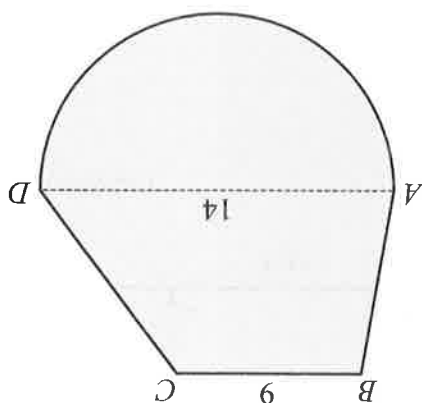


12



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9



This shape is made from a trapezium and a semicircle.

$AD = 14$ cm and $BC = 9$ cm.

The perpendicular height of the trapezium is equal to the radius of the semicircle.

Calculate the shaded area.

13

$$a = \sqrt{3b - c}$$

(a) Find a when $b = 0.08$ and $c = -0.12$.

Answer $a = \dots\dots\dots$ [1]

(b) Rearrange the formula to make b the subject.

Answer $b = \dots\dots\dots$ [2]

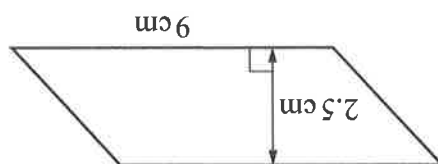


14 (a)



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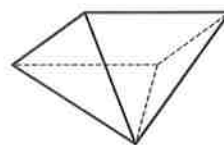
10



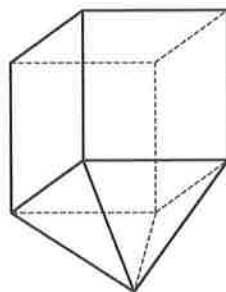
Calculate the area of the parallelogram.

Answer cm² [1]

(b)



Solid A



Solid B

Solid A is a pyramid with a surface area of 85 cm².
Solid B is formed by placing solid A onto a cube of side 6 cm.

Calculate the surface area of solid B.

Answer cm² [2]



[3]

Answer

Show that the radius of the cone is 4.68 cm, correct to 2 decimal places.

- 16 The height of a cone is 2.8 times its radius.
The volume of the cone is 300 cm^3 .

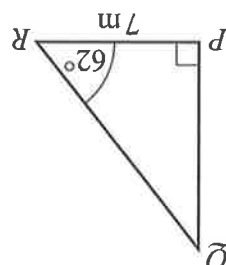
Answer $x = \dots\dots\dots$ [1]

Given that x° is an obtuse angle, find x .

(b) $\cos 68^\circ = -\cos x^\circ$

Answer $\widehat{QR} = \dots\dots\dots \text{m}$ [2]

Calculate $\widehat{QR}$.



15 (a)



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17 A car park contains cars that are grey, red, black or blue. Aiden has drawn this pictogram for the grey, red and black cars in the car park.

Grey	
Red	
Black	
Blue	

Key: represents 4 cars

(a) Find the number of red cars in the car park.

Answer [1]

(b) The total number of cars in the car park is 58.

(i) Aiden has not yet drawn the pictogram to show the number of blue cars in the car park.

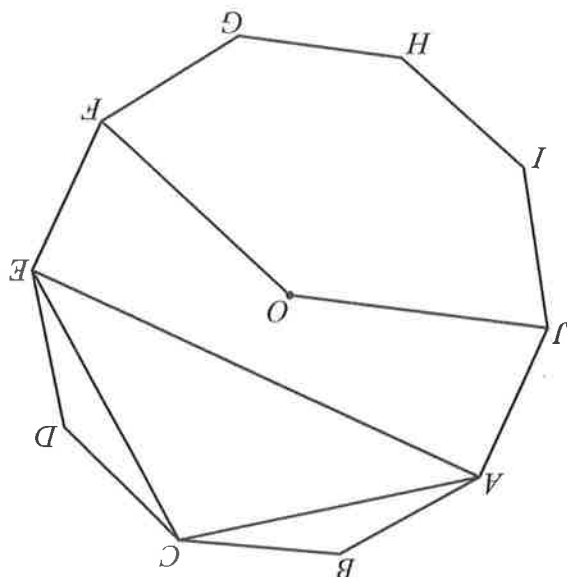
Find the number of blue cars in the car park.

Answer [1]

(ii) Calculate the percentage of the cars in the car park that are grey.

Answer % [2]





The diagram shows a regular decagon, centre O .
The interior angle of any regular decagon is 144° .

(a) Calculate angle BAC .

Answer Angle $BAC = \dots\dots\dots$ [1]

(b) Calculate angle ACE .

Answer Angle $ACE = \dots\dots\dots$ [1]

(c) Calculate obtuse angle FOJ .

Answer Angle $FOJ = \dots\dots\dots$ [1]



19 A is the point $(4, 5)$, B is the point $(10, -2)$ and C is the point $(-2, k)$.

(a) Calculate the length of AB .

(b) The gradient of AC is $-\frac{2}{3}$.

Find the value of k .

Answer $AB = \dots\dots\dots$ units [2]

Answer $k = \dots\dots\dots$ [2]



Answer $p = \dots\dots\dots$ [4]

Calculate p .

The mean number of strawberries per box is 19.75 .

Frequency	25	29	p	19	15
Number of strawberries	18	19	20	21	22

20 Some boxes are filled with strawberries.
The number of strawberries in each box is counted.
The table shows the results.



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21 (a) Solve $5(x-3) = 7x+2$.

(b) Solve these simultaneous equations.

$$\begin{aligned} 3x+4y &= 1 \\ x-2y &= -8 \end{aligned}$$

Show your working.

Answer $x =$ [2]

Answer $x =$

$y =$ [3]

22 (a)



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17

Monthly water usage ($> 40 \text{ m}^3$)	Price per m^3 (\$)
Tariff	1.52
Water Conservation tax (65% of tariff)	x
Waterborne Fee	1.18
Total Price	y

The table shows the price per m^3 of water when more than 40 m^3 of water is used in a month. The total price, in dollars, per m^3 of water is the sum of the tariff, water conservation tax and waterborne fee.

Calculate the value of x and the value of y .

Answer $x =$

$y =$

[2]

(b) A customer uses p kilowatt-hours (kWh) of electricity in one year.

Electricity costs \$0.2743 per kWh.

Goods and Services Tax (GST) is then charged at a rate of 8%.

The total amount the customer pays for the electricity for the year is \$352.93 .

Calculate p .

Answer $p =$

[3]

[Turn over]





Show your working.

23 Solve $\frac{3}{x-2} + \frac{x+1}{8} = 5$.

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Answer $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [5]

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